Chapter 10:
Homeostasis and thermoregulation

Homeostasis introduction

Homeostasis terminology

Homeostasis
The processes involved in maintaining a constant internal environment, within tolerance limits, despite changes in the internal and external environment

Stimulus–response model
Two stages of homeostatic regulation: detection of a change from the stable state, known as the stimulus; response to the stimulus, described as counteracting the change

Biochemical pathway
A linked series of chemical reactions in living organisms

Metabolism
The sum total of the chemical reactions in biochemical pathways in organisms

Enzymes
Reusable, biological catalysts (proteins) that lower the activation energy of a chemical reaction, enabling it to proceed faster

Detecting stimuli: receptors

There are millions of external and internal receptors that allow an organism to respond to stimuli.

Type	Stimuli	Location in animals
Chemoreceptor	(Internal receptors) Detect oxygen and ion levels	Aorta, carotid arteries
Osmoreceptor	(Internal receptors) Detect changes in osmotic pressure in blood (changes in blood water concentration)	Hypothalamus
Photoreceptor	(External receptors) Light	Eyes; light-sensitive cells on body surface of some invertebrates
Thermoreceptor	External receptors) Detect external temperature variations	Skin
	Internal receptors) Detect internal temperature variations	Hypothalamus

The nervous system: effectors and neurons

The nervous system comprises the central nervous system (CNS) and the peripheral nervous system (PNS). The brain and the spinal cord form the CNS.

Nerve impulses travel along defined pathways. They follow the **sensory neurons** from the source of stimulation, via the PNS to the CNS. **Interconnecting neurons** located in the CNS relay the electrical impulses from the sensory neurons to the appropriate **motor neurons**. From the CNS, motor neurons relay signals via the PNS to the effectors along different pathways.

Effectors are the muscles or glands that respond to the stimuli.

The endocrine system

Not all changes in an organism's internal and external environment require an immediate response. Some responses take time and are under the control of hormones produced by the endocrine system.

Endocrine gland	Hormone secreted	Target tissue/ organ	Function related to homeostasis
Posterior pituitary	Antidiuretic hormone	Kidney	Stimulates reabsorption of water
Thyroid	Thyroxine	Nearly all tissues	Increases metabolic rate, and as a result increases oxygen consumption and production of heat

The nervous and endocrine systems work together during homeostasis. The Nervous system executes a relatively fast response along neurons and the endocrine system executes a relatively slow response.

Negative feedback loop/model

A **negative feedback loop** (or model) is a homeostatic process that changes the direction of a stimulus The stimulus initially deviates away from normal. A negative feedback loop will always reduce the stimulus.

Negative feedback can be modelled using a flow diagram. The negative feedback model is represented as a cycle but, in reality, the mechanism is only cyclic if a stimulus continuously occurs after negative feedback.

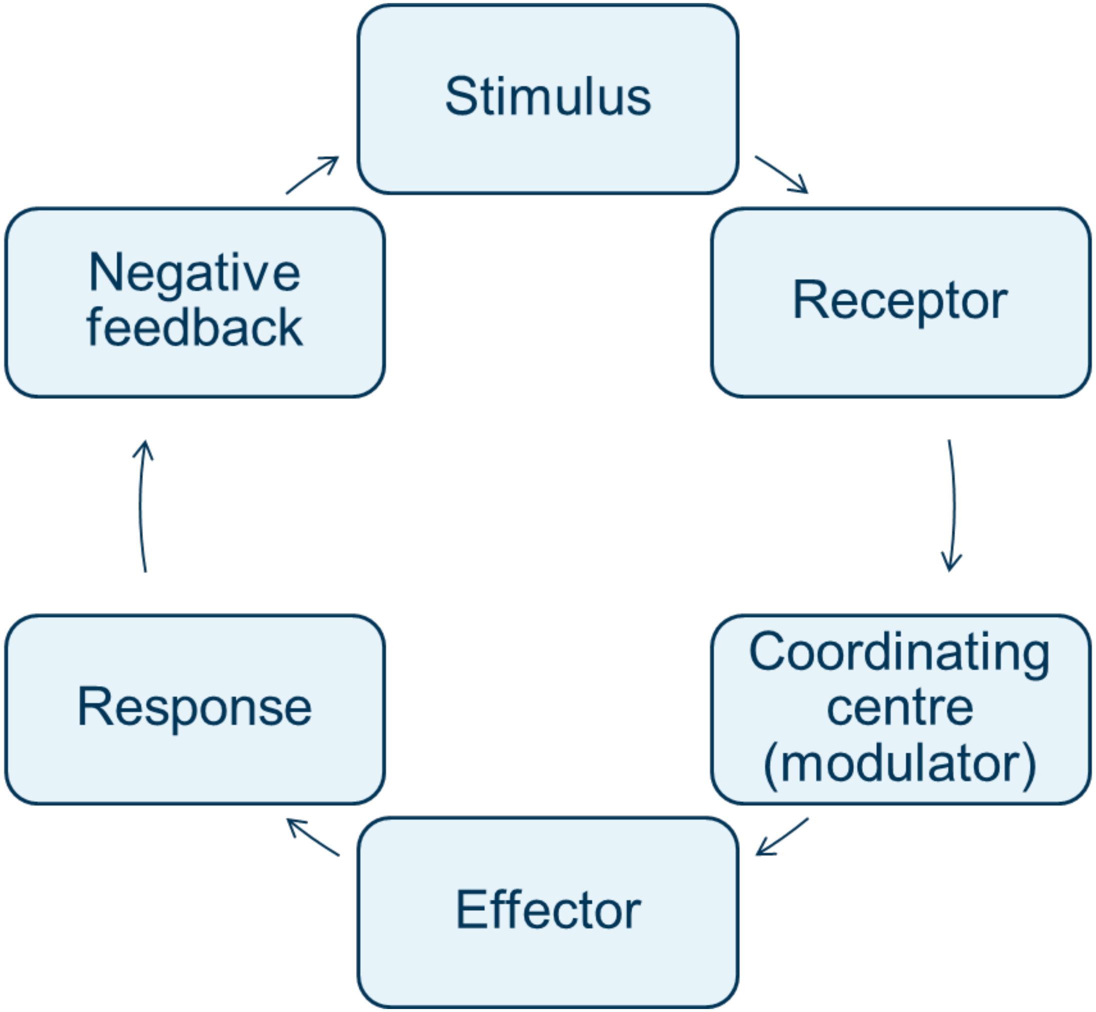
Negative feedback

Negative feedback loop/model

Part of model	What it is and what it does
Stimulus	A change in one of the internal or external environmental factors that is detected by a receptor. The change involves a deviation away from the normal/optimal value.
Receptor	The cell or tissue that detects the stimulus (change in the environment). The receptor may be internal or external.
Coordinating centre (modulator)	(Usually the hypothalamus) receives messages from receptors (via sensory neurons), coordinates a response, then sends an instruction to an effector via motor neurons.
Effector	A *muscle* or *gland* that receives the message from the control centre and carries out the response
Response	The action of the effector that counteracts the stimulus (i.e. the change made by the effector)
Negative feedback	A message that counteracts the stimulus; it returns the value back to its normal/ optimal value

Negative feedback

Negative feedback loop/model



Tolerance ranges

Each organism has set ranges within which they can tolerate the temperature, water balance and different levels of organic and inorganic materials.

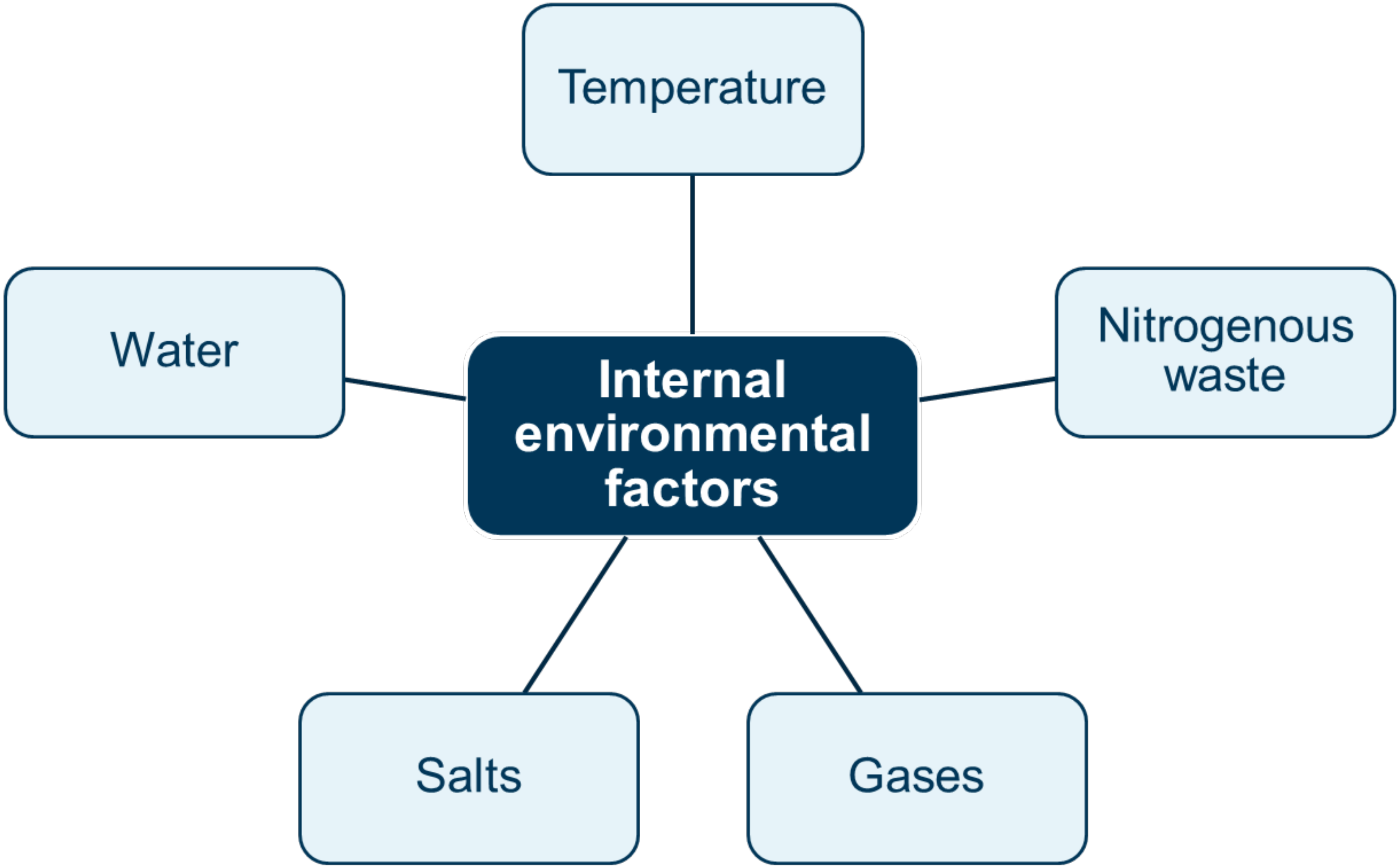
These are known as **tolerance ranges**.

Organisms can survive change in a factor if it remains within the tolerance range for that factor.

If an internal environmental factor goes outside of the tolerance range, it may be fatal for the organism.

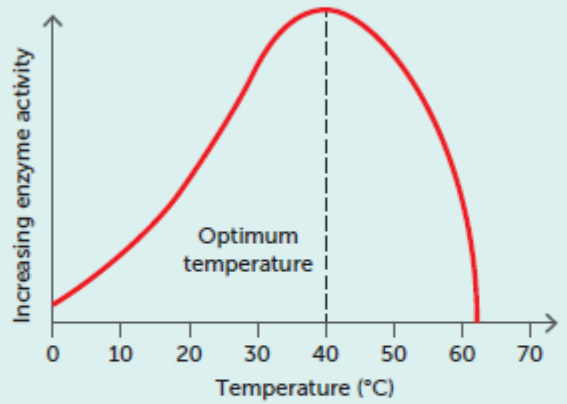
Tolerance limits

Internal environmental factors



Tolerance limits

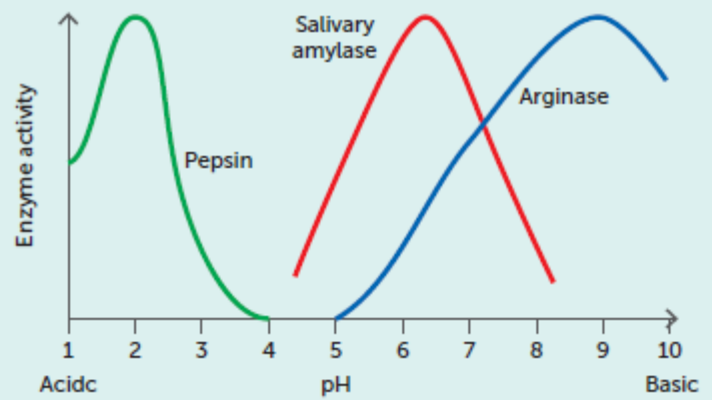
Internal environmental factors: temperature



As temperature increases so does the rate of reaction. However, at high temperatures, some of the bonds that hold the enzyme together break, changing the shape of the active site so the enzyme becomes denatured.

Tolerance limits

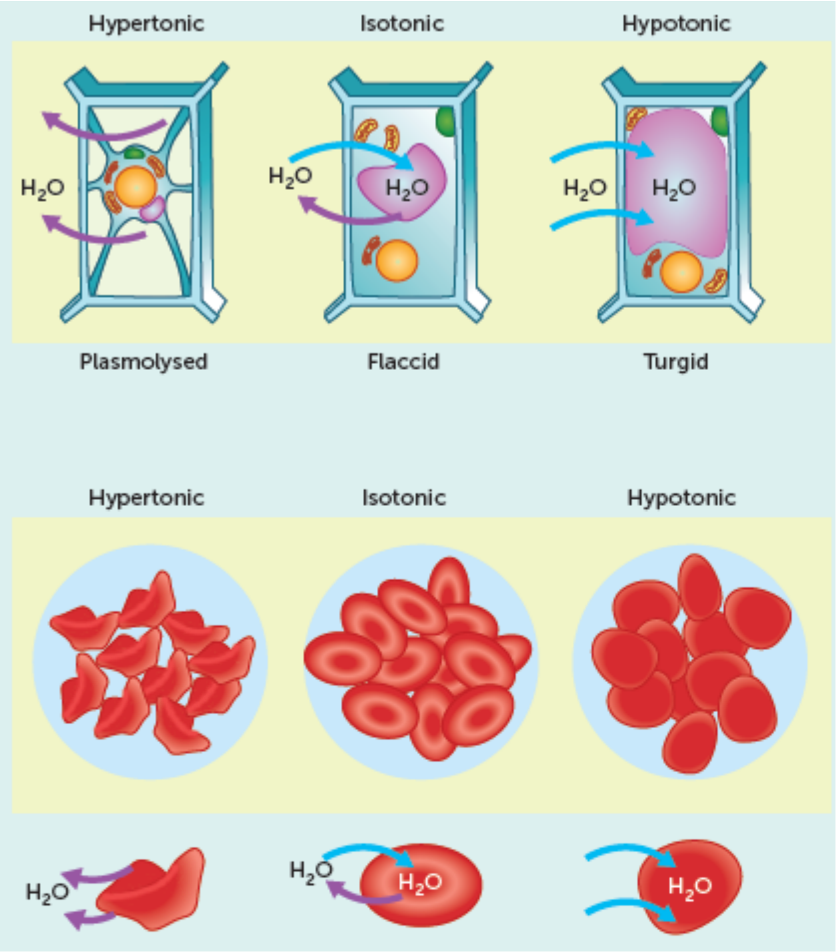
Internal environmental factors: nitrogenous waste



Different enzymes work best at different pHs. If the pH changes too far from this optimum, the bonds holding the enzyme together weaken, changing the shape of the active site so the enzyme becomes denatured.

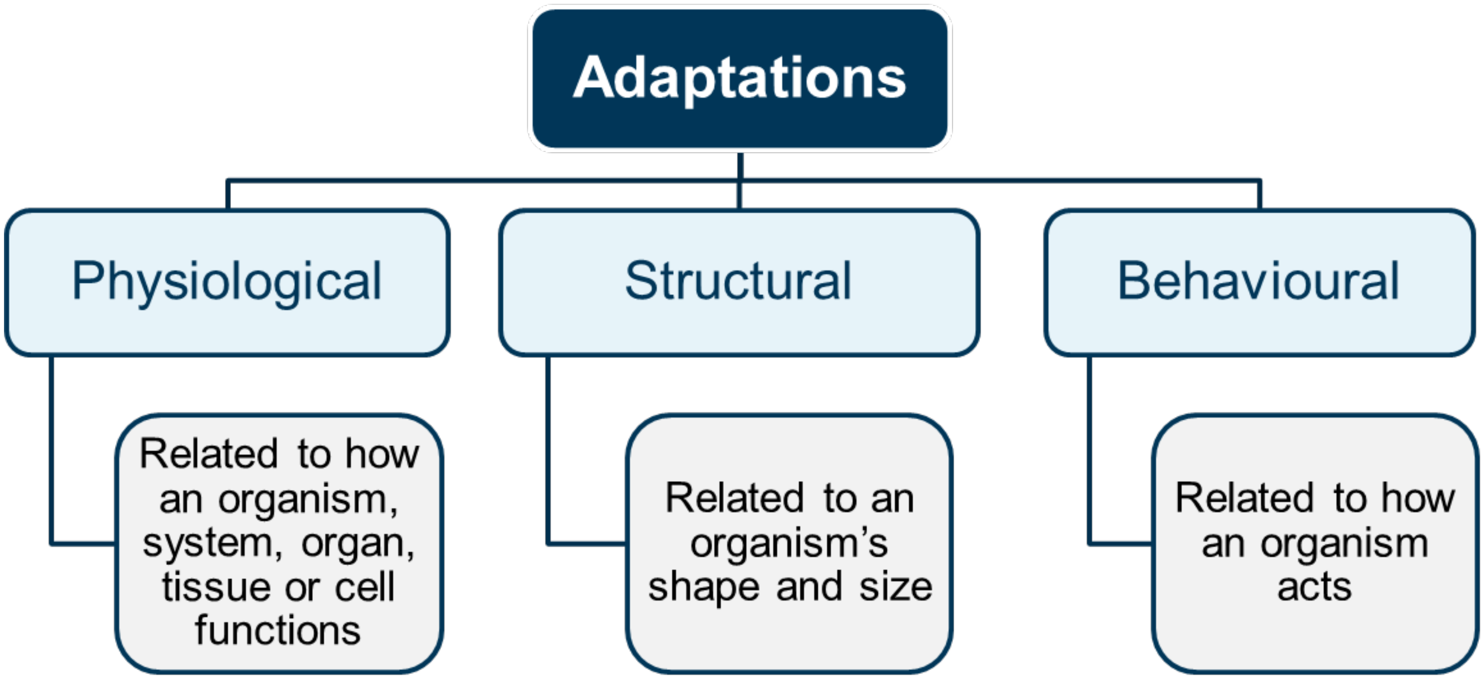
Tolerance limits

Internal environmental factors: water



What are adaptations?

Adaptations are features and strategies used by organisms to help them survive in a particular environment.



Adaptations for survival

Behavioural adaptation example:

The moist, delicate skin of a frog can dry out easily, preventing it from achieving adequate gas exchange and leading to death. The desert spadefoot frog's behaviour of burrowing underground during the dry months prevents exposure to the sun while it waits safely for rainfall.

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Thermoregulation

Most individual species can only survive within a relatively narrow range of temperature, and they cannot exist in habitats that have greatly varying temperatures. Within their tolerance range, each species has an optimal temperature range at which it functions best.

Thermoregulation is critical for survival, because most biochemical and physiological processes are temperature sensitive.

Thermoregulatory mechanisms include structural features, behavioural responses and physiological mechanisms to control heat exchange and metabolic activity.

Thermoregulation

Thermoregulation in plants

Australian snow gum (*Eucalyptus pauciflora*) and *Hakea coriaceae* use mechanisms to control heat exchange and metabolic activity.

***Eucalyptus pauciflora* (cold climates)**	***Hakea coriaceae* (hot climates)**
Thick, leathery, waxy leaves reduce heat loss by providing insulation.	Narrow, vertical leaves minimise the amount of direct sunlight hitting them, and thus heat absorption. They also increase resistance to cold temperatures and frosts, which deserts can experience at night.

Thermoregulation in animals

Thermoregulation in animals is the process by which they maintain an internal temperature within a tolerance range. Heat for thermoregulation can either come from metabolism or from the external environment.

Animals that use metabolic processes to generate their own heat to maintain their internal temperature within the tolerance range are called **endotherms**. Animals such as birds and mammals can generally maintain a stable internal temperature independent of external temperature fluctuations.

An animal whose body temperature is determined by the external environment is called an **ectotherm**. Animals such as amphibians, some reptiles and fish, and most invertebrates, cannot maintain a stable internal environment and instead obtain heat from the sun or from objects in their surroundings.

Thermoregulation

Endotherms and ectotherms compared

Endotherms	Ectotherms
Use metabolic processes to generate their own heat to maintain their internal temperature	Rely on external sources for gaining heat; may obtain heat from the sun or from objects in the surroundings
Maintain a stable internal temperature independent of external temperature fluctuations	Body temperature fluctuates with the external environment
Birds and mammals	Most amphibians and reptiles

Costs and benefits of endothermic and ectothermic mechanisms

	Endotherms	Ectotherms
Cost	To maintain stable internal temperature, they may have higher metabolic rate Need to spend more energy to maintain a higher metabolic rate Results in higher food requirements and more time spent finding food	Body temperature dependent on the external environment Limited to living in environments with less extreme temperatures Cannot tolerate very high or very low external temperatures
Benefit	Body temperature independent of external temperature Can live in extreme environments Can be active at night or more often during the day and in cold weather Being more active may reduce the chance of predation	Heat source is mainly the environment, so there are lower energy requirements Do not need to consume as much food Can spend less time hunting for food Can tolerate large fluctuations in internal boy temperature

Adaptations for thermoregulation

Heat transfer

Heat transfer depends on the temperature gradient between the internal and external environments.

When there is a balance between heat gain and heat loss, the organism is said to be in heat balance, which is the purpose of thermoregulation.

An organism that is hotter than its surroundings loses heat energy in four ways, and an organism that is cooler than its surroundings gains heat in the same four ways: **conduction**, **convection**, **evaporation** and **radiation**.

Adaptations for thermoregulation

Types of heat transfer

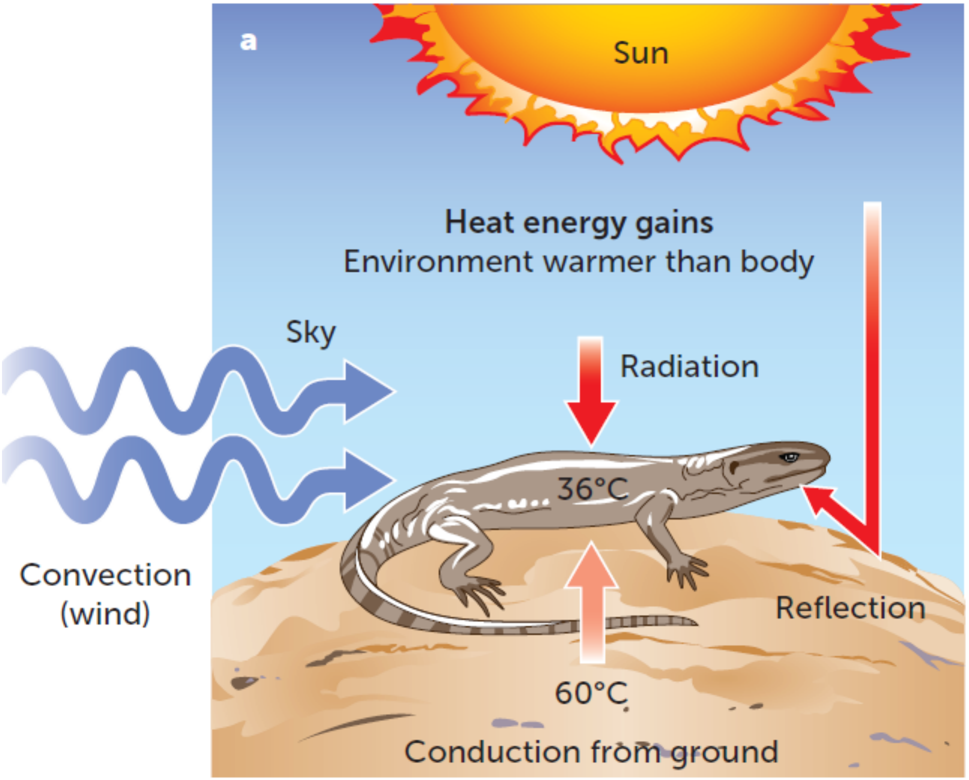
Types of heat transfer

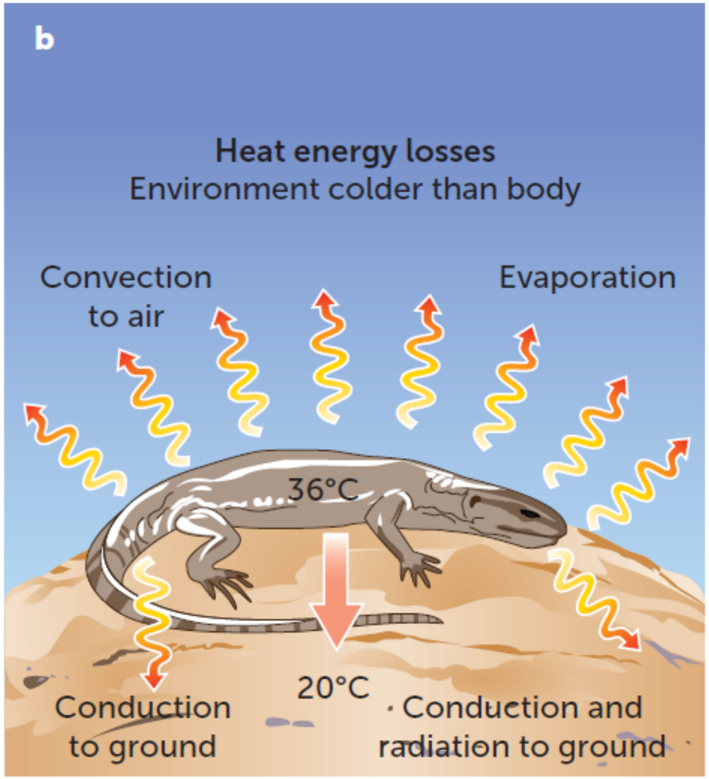
- **Conduction**
  - Hotter object to cooler object by direct contact
- **Convection**
  - Currents of air remove heat energy from surface of organism
- **Evaporation**
  - Water turns to water vapour, absorbing heat
- **Radiation**
  - Emission of electromagnetic ‘heat’ waves

Adaptations for thermoregulation

Heat transfer example 1:

Lizards are ectothermic, so rely on external heat sources. Lizards use a number of heat transfer methods to regulate their temperature during the day and night.





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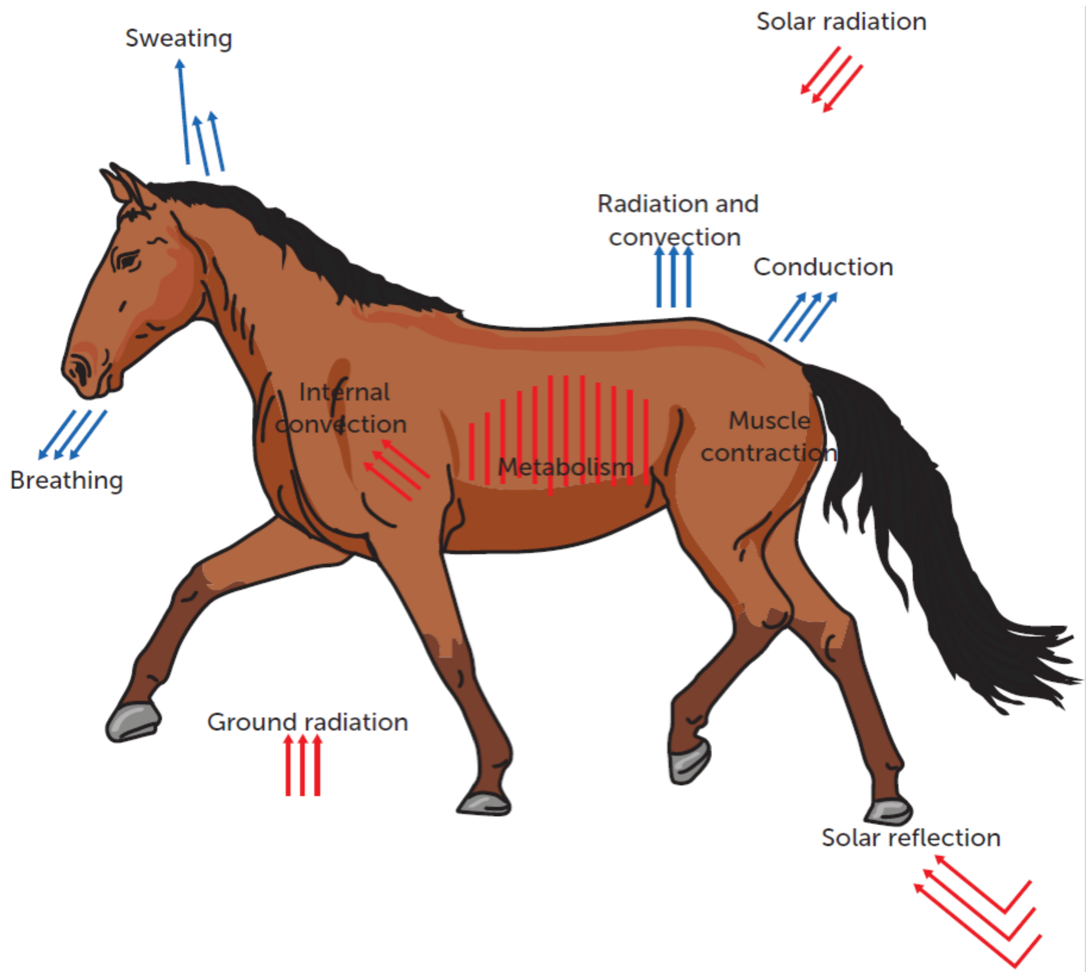
Adaptations for thermoregulation

Heat transfer example 2:

Horses are endothermic, so can regulate their body temperature independent of the external temperature. Horses use a number of heat transfer methods to regulate their temperature while still and while active.



Adaptive structural features in endotherms

Structural features can help cool endotherms when needed.

Example:

Elephants do not have sweat glands. Instead, when their surroundings are hot, elephants increase the blood supply to their ears and flap them around to lose body heat. The efficiency of this mechanism is enhanced by the high surface area-to-volume ratio of their large flat ears. They can achieve a high rate of heat transfer from the veins to the surroundings via radiation.

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Adaptations for thermoregulation

Adaptive structural features in ectotherms

Ectotherms have some structural features that can be used in hot weather.

Example:

The frilled-neck lizard's skin colour can be adjusted to keep its internal temperature within the tolerance range. When desert temperatures rise, their colour becomes lighter, which reflects the heat and keeps the lizard cooler.

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Adaptations for thermoregulation

Adaptive behavioural responses

To reduce heat gain, some animals shelter from high temperatures and reduce their activity, only emerging to feed in the relative cool of dusk and dawn. Avoiding daytime sun, such as by resting in the shade, reduces heat gain via radiation from the sun and via conduction from hot objects such as rocks.

Examples:

Red kangaroos lick their wrists where the blood vessels form a dense network close to the surface. The evaporation has a cooling effect.

The burrow is a refuge for meerkats because it stays within a narrower temperature range than the air outside, and it is suited to their tolerance range.

Physiological adaptations for surviving in hot environments

Sweat glands open to release water and salt onto the skin, which evaporates and cools the skin. This is known a **sweating**. The evaporated water carries away the heat energy from the body and lowers the internal temperature. This is evaporative cooling.

Many animals, such as dogs, **pant** to keep cool. Panting expels hot air and brings in cooler air, which then helps moisture in the mouth to evaporate quickly, reducing body temperature.

Another response used by endotherms is a decrease in the **metabolic rate**, which reduces the amount of heat generated within the body.

Finally, the muscles attached to hair follicles can be relaxed to flatten the hairs and decrease the layer of air acting as an insulator. This is known as **pilorelaxation**.

Adaptations for thermoregulation

Structural adaptations for surviving in cold environments

For animals living in low-temperature environments, the problem is how to reduce heat loss and increase heat conservation. Several structural features allow them to do this.

Structural features in endotherms and ectotherms for surviving the cold

- **Insulation**
  - Surface and internal insulation reduces heat flow from organism to environment
- **Piloerection**
  - Raising fur or feathers
- **Countercurrent heat exchange**
  - Exchange of heat between two fluids (blood) flowing in opposite directions
- **Surface area-to-volume ratio**
  - Lower surface area-to-volume ratio means less heat lost

Adaptations for thermoregulation

Structural adaptation to cold example:

Adaptations that reduce the surface area-to-volume ratio reduce heat loss, because there is less surface area per unit volume for heat to transfer through. Small extremities such as ears and legs can also reduce the rate of heat loss. The ears and limbs of Arctic foxes are more rounded and smaller than those of their relatives the red fox and the grey fox.

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Adaptations for thermoregulation

Behavioural adaptations for surviving in cold environments

Animals can reduce heat loss by reducing the amount of surface area exposed to their surroundings. Several behaviours aid in this.

Behavioural features in endotherms and ectotherms for surviving the cold

- Huddling
  - Reduces group's overall surface area-to-volume ratio
- Basking in sun
  - Gains heat via radiation
- Lying on hot rock
  - Gains heat by conduction
- Lying flat to increase body surface area
  - Maximises heat gain by radiation and conduction

Adaptations for thermoregulation

Physiological adaptations for surviving in cold environments

Vasoconstriction is a mechanism for heating the body and counteracts a decrease in internal body temperature. A negative feedback model represents the mechanism at work.

Part of model	What it is and what it does
Stimulus	Decrease in external environmental temperature below optimal range
Receptor	Thermoreceptor cells in the skin detect the change in external temperature and send a message to the coordinating centre.
Coordinating centre (modulator)	The hypothalamus receives a message from the receptor, coordinates a response and sends a message to the effector.
Effector	Smooth muscle cells within the walls of the blood vessels contract, reducing the diameter of the arterioles.
Response	Vasoconstriction – reduced blood flow in the vessels close to the surface of the skin – reduces heat loss via radiation.
Negative feedback	Internal temperature returns to normal/optimal value. The mechanism counteracted the stimulus.